

The Intonational Phrase

7.0. Introduction

Now that the phonological phrase has been constructed, we can construct the next unit in the prosodic hierarchy, the intonational phrase (*I*). *I* groups together one or more ϕ s on the basis of syntactic information, but as will be seen below, the nature of this information is more general than that needed for the definition of the ϕ domain. This shows clearly how the higher a constituent is in the prosodic hierarchy, the more general the nature of its definition becomes. In addition to the basic syntactic factors that play a role in the formation of intonational phrases, there are also semantic factors related to prominence and performance factors such as rate of speech and style that may affect the number of intonation contours contained in an utterance. Thus, any definition of the *I* domain must allow for this variability. While the relatively large degree of variability in the organization of a string into *I*s distinguishes *I* from the other prosodic constituents, it will be shown in this chapter that there are nevertheless syntactic and semantic constraints on what may constitute an intonational phrase. It will be shown, furthermore, that the strings delimited by the constituent *I* not only account for ‘the domain over which an intonational contour is “spread”’ (Selkirk, 1978b:130), but they also represent the domain of application of a number of segmental phonological rules. Finally, it should be noted that while intonation contours are essential in the discussion of the *I* constituent of the phonological hierarchy, we will limit our discussion of these contours to their domains. We will not, therefore, discuss the actual phonetic patterns of the contours.¹

7.1. Definition of the intonational phrase domain

In this section, we will discuss the basic rule for *I* formation. As will be seen, this rule uniquely partitions a sequence of ϕ s into *I*s, and as it stands, does not allow for any variability in *I* structure. The variability that is characteristic of the intonational phrase level in the prosodic hierarchy will be treated in the next section, where it will be proposed that it is the

result of a separate type of rule, that is, a restructuring rule.

The formulation of the basic *I* formation rule is based on the notions that the intonational phrase is the domain of an intonation contour and that the ends of intonational phrases coincide with the positions in which pauses may be introduced in a sentence.² It will be demonstrated below, however, that the domains established on the basis of intonation contours and the potential for pauses are the same as those needed to account for the application of a number of segmental rules in several languages.

One of the first things that is usually observed in relation to intonation contours is that there are certain types of constructions that seem to form intonation domains on their own. These constructions include parenthetical expressions, nonrestrictive relative clauses, tag questions, vocatives, expletives, and certain moved elements (see, among others, Bing, 1979; Downing, 1970; Ladd, 1980; Selkirk, 1978b, 1984b).³ Examples of each of these structures are given in (1). Although the examples in (1) are all in English, it is our proposal that the same types of constructions will obligatorily form *I*s in all languages that make use of them.

- (1)
- a. Lions [_Ias you know]_I are dangerous.
 - b. My brother [_Iwho absolutely loves animals]_I just bought himself an exotic tropical bird.
 - c. That's Theodore's cat [_Iisn't it?]_I
 - d. [_IClarence]_I I'd like you to meet Mr. Smith.
 - e. [_IGood heavens]_I there's a bear in the back yard.
 - f. They are so cute [_Ithose Australian koalas]_I.

While the different types of constructions that obligatorily form *I*s in some sense appear to constitute a rather disparate set of items, in another sense they all share a common property. That is, they all represent strings that are in some way external to the root sentence they are associated with. Following Safir's (1985) recent proposal in relation to nonrestrictive relative clauses, it seems that we might be able to account for the set of obligatory *I*-forming constructions on the grounds that they are all elements that, at the level of s-structure, are linearly represented but not structurally attached to the sentence tree.

It should be noted, in addition, that those types of constructions that obligatorily form an *I* do so regardless of where they occur in a sentence, as illustrated in (2).

- (2)
- a. [_IAs you know]_I Isabelle is an artist.
 - b. Isabelle [_Ias you know]_I is an artist.
 - c. Isabelle is [_Ias you know]_I an artist.
 - d. Isabelle is an artist [_Ias you know]_I.

Aside from the specific types of constructions that obligatorily form *Is*, the other syntactic notion that is relevant for *I* formation is the root sentence, as defined by Emonds (1976). Specifically, the boundaries of a root sentence delimit an *I*, while those sentences that are not root sentences do not. This is clearly illustrated by the following pair of sentences given by Downing (1970):

- (3) a. [_IBilly thought his father was a merchant]_I [_Iand his father was a secret agent]_I
 b. [_IBilly thought his father was a merchant and his mother was a secret agent]_I

It should be noted that there are also cases in which a root sentence does not form a single *I*, and that is when it is interrupted by one of the types of constructions seen above that must obligatorily form an *I* on its own. Such cases have, in fact, already been seen in (1b), (2b), and (2c). Let us consider the case of the root sentence *Isabelle is an artist*, which in (2b) and (2c) is interrupted by the parenthetical element *as you know*. Since the parenthetical must form an *I* by itself, it is no longer possible for *Isabelle is an artist* to form a single *I*. In such cases, any string that is adjacent to the obligatory *I*, and which would not form an *I* for independent reasons, automatically forms another *I* on its own.⁴ Thus, the parts of the sentence *Isabelle is an artist* that precede and follow *as you know* form separate *Is*, as seen below, although they are neither the types of constructions that obligatorily form *Is* nor are they root sentences.

- (4) a. [_IIsabelle]_I [_Ias you know]_I [_Iis an artist]_I
 b. [_IIsabelle is]_I [_Ias you know]_I [_Ian artist]_I

On the basis of the different criteria seen above for delimiting *Is*, it is possible to formulate the basic definition of *I* as follows:

(5) *Intonational Phrase Formation*

I. *I domain*

An *I* domain may consist of

- a. all the ϕ s in a string that is not structurally attached to the sentence tree at the level of s-structure, or
 b. any remaining sequence of adjacent ϕ s in a root sentence.

II. *I construction*

Join into an n-ary branching *I* all ϕ s included in a string delimited by the definition of the domain of *I*.

It should be noted that while the *I* domain often corresponds to a syntactic constituent, this is not always the case. As expressed in (5), *I* is isomorphic with any of the constituent types that obligatorily form intonational phrases and with a root sentence if it is not interrupted by an obligatory *I*. When a root sentence does have an intervening obligatory *I*, however, it is often the case that the strings on one or both sides of this *I* are not isomorphic to any constituent in syntax. This was seen in the example in (4b), where *Isabelle is* does not form a syntactic constituent,⁵ and it is further illustrated in the examples in (6).

- (6) a. [_IThey have]_I [_Ias you know]_I [_Ibeen living together for years]_I
 b. [_IHe will never]_I [_Ias I said]_I [_Iaccept your proposal]_I
 c. [_ICharles wouldn't]_I [_II imagine]_I [_Ihave done such a thing]_I
 d. [_IThat's the tortoise that]_I [_Ias you know]_I [_Iinhabits the Galapagos Islands]_I

In all of these examples, the parenthetical expression forms an obligatory *I* internal to the root sentence within which it is placed, and the strings on either side of this *I* must thus also form *Is*. The result is that the peripheral *Is* delimit strings that are not necessarily isomorphic to any constituent in syntactic structure. While in right recursive languages such as English, it seems that *Is* that are not isomorphic to a syntactic constituent may appear only to the left of the obligatory *I*, it may be that in other types of languages other possibilities also exist. Thus, as in the case of the other prosodic constituents seen so far, *I* makes use of (morpho-)syntactic notions in its definition, though the resulting structures are not necessarily in a one-to-one relation to any constituent found elsewhere in the grammar.

As far as the relative prominence of the ϕ s within an *I* is concerned, the position of the strong node cannot be determined for the category as a whole, as it is in the case of the other prosodic constituents. That is, while for any of the constituents below *I* it was possible to specify the position in which the strong sister node was located, at the *I* level we must allow the strong position to vary. Rather than being determined structurally, the strong node under *I* is determined on the basis of semantic factors such as focus and given *vs.* new information. The information needed for determining which is the strong element may be present in the sentence itself, such as the use of an indefinite article (indicating new information) as opposed to a definite article (indicating old information), as illustrated in (7).

- (7) a. $[_I[\text{Leonard}]_{\phi_w} [\text{found}]_{\phi_w} [\text{a package}]_{\phi_s} [\text{on the doorstep}]_{\phi_w}]_I$
 b. $[_I[\text{Leonard}]_{\phi_w} [\text{found}]_{\phi_w} [\text{the package}]_{\phi_w} [\text{on the doorstep}]_{\phi_s}]_I$

In many cases, however, the assignment of stress within an intonational phrase depends on material found in previous utterances or on shared knowledge of a given situation that is not necessarily present in the linguistic context of the utterance in question. On the basis of these observations, a maximally general rule for the relative prominence relations within *I* can be formulated as follows:

(8) *Intonational Phrase Relative Prominence*

Within *I*, a node is labeled *s* on the basis of its semantic prominence; all other nodes are labeled *w*.

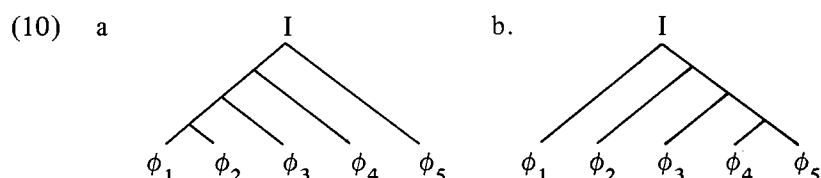
This rule allows for a certain degree of flexibility in stress placement within *I* which is absent at the other levels of the prosodic hierarchy. For example, any of the ϕ s in the sentence *My sister sells fresh fruit at the market on Monday* could potentially be labeled *s*, as shown in (9), depending on which element is the most prominent semantically or represents the focused element of the sentence.

- (9) a. $[_I[\text{My sister}]_{\phi_s} [\text{sells}]_{\phi_w} [\text{fresh fruit}]_{\phi_w} [\text{at the market}]_{\phi_w} [\text{on Monday}]_{\phi_w}]_I$
 b. $[_I[\text{My sister}]_{\phi_w} [\text{sells}]_{\phi_s} [\text{fresh fruit}]_{\phi_w} [\text{at the market}]_{\phi_w} [\text{on Monday}]_{\phi_w}]_I$
 c. $[_I[\text{My sister}]_{\phi_w} [\text{sells}]_{\phi_w} [\text{fresh fruit}]_{\phi_s} [\text{at the market}]_{\phi_w} [\text{on Monday}]_{\phi_w}]_I$
 d. $[_I[\text{My sister}]_{\phi_w} [\text{sells}]_{\phi_w} [\text{fresh fruit}]_{\phi_w} [\text{at the market}]_{\phi_s} [\text{on Monday}]_{\phi_w}]_I$
 e. $[_I[\text{My sister}]_{\phi_w} [\text{sells}]_{\phi_w} [\text{fresh fruit}]_{\phi_w} [\text{at the market}]_{\phi_w} [\text{on Monday}]_{\phi_s}]_I$

While (8) is formulated in a maximally general way, and all of the pronunciations in (9) are possible, the assignment of relative prominence within *I* is not totally free; there are certain possibilities that are definitely

marked with respect to others. For example, of the patterns seen in (9), the first two represent contrastive patterns and thus seem to be marked with respect to the last one, which is the least marked pattern. A more refined version of the *I*-level Relative Prominence rule will undoubtedly also have to account for such differences in markedness.

It should be noted in relation to the flexibility in the position of the strong element within *I*, furthermore, that the *n*-ary branching prosodic trees proposed in this book offer several advantages over the binary branching trees proposed in earlier work on prosodic phonology (Selkirk, 1978b, 1980a; Nespor and Vogel, 1982). In phonological theories with binary branching trees, the assignment of relative prominence is generally coupled with the geometry of the tree itself. That is, the position of the stressed element in a binary branching tree is determined by the direction of branching of the tree and by a set of principles that specify *a*) whether in a pair of sister nodes $N_1 N_2$, N_1 or N_2 is strong, and *b*) whether or not this choice depends on the branchingness of the nodes. Such principles greatly reduce the number of possible positions for the strong element in a given string; however, in the case of *I*, this is not a desirable result, since there are instances in which the *s* falls on a node that could not be so labeled by any of the generally accepted relative prominence labeling conventions. Consider the abstract sequences of five ϕ s grouped into left and right branching *I*s in (10a) and (10b), respectively.



In a left branching tree, the usual labeling conventions can either place *s* on ϕ_1 (i.e., given $N_1 N_2$, N_1 is *s*) or on ϕ_2 (i.e., given $N_1 N_2$, N_1 is *s* iff it branches). In a right branching tree, the labeling conventions will place *s* either on ϕ_5 (i.e., given $N_1 N_2$, N_2 is *s*) or on ϕ_4 (i.e., given $N_1 N_2$, N_2 is *s* iff it branches). There is no way, however, to label ϕ_3 strong. Thus, the labeling conventions would not be able to handle the type of pronunciation in (9c) of the sentence *My sister sells fresh fruit at the market on Monday*. More generally, it would be impossible to assign stress to any node other than the two leftmost or the two rightmost ones. Such a restriction is not built into the relative prominence conventions for *n*-ary branching trees, and in the case of *I* in particular, this is a desirable outcome since there are clearly cases in which the strong node can occupy a position other than one of the four predictable on the basis of binary tree structures. A further advantage of *n*-ary branching trees also

derives from the fact that there is no theoretical connection between the tree geometry and the *s/w* assignment rules. It should be noted that the absence of such a connection does not mean that there is no way to restrict the assignment of stress in such trees; in fact, for all of the prosodic constituents below *I*, it was shown that stress always falls on a specific position within a given type of constituent and is thus restricted structurally. In the case of *I*, the restrictions are more semantic in nature, but they nevertheless exist, since it is clearly not the case that the position of stress is random. In a system in which the position of stress may follow only from the structure of the tree, however, any placement of stress on a position other than the predicted one would constitute a violation of the principles of the system. In such a system, the fact that the strong node within an *I* can be in any position and can vary with different pronunciations of the *I* in question would thus mean that in a large number of cases the relative prominence patterns are in conflict with the basic stress assignment principles. In an *n*-ary branching system, where the assignment of stress is not necessarily determined strictly on the basis of the tree structure, no such problem arises. Still another advantage of *n*-ary branching trees will be seen in the next section in relation to the restructuring of *I*.

7.2. Restructuring of the intonational phrase

While constituents of a particular type are always assigned the same structure in syntax regardless of their length or any other linguistic or extra-linguistic factors, the same is not true in phonology. We have already seen in relation to the phonological phrase that, at least in some languages, the length of certain phonological constituents plays a role in determining the ultimate division of a string into ϕ s. In particular, we saw that in languages such as Italian a short (i.e. nonbranching) ϕ may, under certain circumstances, be restructured to form a single, larger phonological phrase with an adjacent ϕ . It will be shown in this section that the intonational phrase, too, may undergo a process of restructuring.⁶ In the case of *I*, however, length is only one of several factors that may play a role in determining restructuring. Three other factors we will also examine here are rate of speech, style, and contrastive prominence. Since *I* restructuring depends in large part on rather global aspects of the speech situation, it is not possible to predict exactly when it will occur. The following discussion is aimed, therefore, at examining the roles the various factors play in increasing or decreasing the probability of restructuring.

Since the basic definition of *I* specifies that all the ϕ s of a root sentence form a single *I*, if the material dominated by a root sentence is long, the resulting *I* is automatically long as well. In such cases, *I* restructuring often occurs to yield somewhat shorter constituents, perhaps for physio-

logical reasons having to do with breath capacity and for reasons related to the optimal chunks for linguistic processing. Thus, in a situation in which no special considerations relating to rate, style, or emphasis must be made, it is possible that a long *I* such as that in (11a) will be broken down into shorter *I*s as in (11b) or (11c).

- (11) a. [_IMy friend's baby hamster always looks for food in the corners of its cage]_I
 b. [_IMy friend's baby hamster]_I [_Ialways looks for food in the corners of its cage]_I
 c. [_IMy friend's baby hamster]_I [_Ialways looks for food]_I [_Iin the corners of its cage]_I

While it may be structurally possible to break down a given *I* into smaller *I*s, it is clear that the longer the original *I*, the more likely it is to be divided into smaller *I*s. This can be seen by comparing the divisions in (11) with those in (12), where the original *I* is already fairly short.

- (12) a. [_IThe hamster eats seeds all day]_I
 b. [_IThe hamster]_I [_Ieats seeds all day]_I
 c. [_IThe hamster]_I [_Ieats seeds]_I [_Iall day]_I

While (12b) and (12c) may represent possible divisions of the sentence into *I*s, they seem much less acceptable than the analogous structures in (11b) and (11c).

The length factor also plays a role in determining how many smaller *I*s are made from a longer one. While it has been suggested that an *I* may in principle be broken down into as many shorter *I*s as there are ϕ s in the string (see Selkirk, 1978b), this does not, in fact, seem to be the case. Aside from the syntactic restrictions on *I* restructuring discussed below, there seems to be a tendency to avoid series of very short *I*s and sequences of *I*s of very different lengths. In other words, there is a tendency to establish *I*s of a more or less uniform, 'average' length, although at this point we are not able to characterize this ideal length more precisely. The importance of length can nevertheless be seen in the examples in (13). While the *I* in (13a) could, in principle, be divided into smaller *I*s as in (13b) and (13c), the divisions in (13d) and (13e) are more acceptable.

- (13) a. [_I[Jennifer] _{ϕ} [discovered] _{ϕ} [that her attic] _{ϕ} [had been invaded] _{ϕ} [last winter] _{ϕ} [by a family] _{ϕ} [of squirrels] _{ϕ}]_I
 b. [_I[Jennifer] _{ϕ}]_I [_I[discovered] _{ϕ}]_I [_I[that her attic] _{ϕ}]_I [_I[had been invaded] _{ϕ}]_I [_I[last winter] _{ϕ}]_I [_I[by a family] _{ϕ}]_I [_I[of squirrels] _{ϕ}]_I

- Fairly closely associated with the factor of length in determining the restructuring of a given *I* into smaller *I*s is the factor of the rate of speech. The more quickly a string is uttered, the less likely it is to be broken down into several intonational phrases. Thus, the faster the rate of speech, the longer the *I*s of a given utterance tend to be; conversely, the slower the rate of speech, the shorter the *I*s tend to be. In this way, some more abstract notion of length in terms of timing or rhythm may be respected. That is, in terms of timing, it may be that a quickly uttered long *I* is in some sense equivalent to a more slowly uttered short *I*. Thus, for example, if the sentence in (11) above is uttered at a fairly rapid tempo, it will typically contain a single long *I*, as in (11a). If it is uttered more slowly, however, it is more likely that the same sentence will contain two or three shorter *I*s, as in (11b) and (11c), respectively.

(14) a. [_IThe adult orangutan constructs a nest every evening out of
leaves and twigs]_I
b. [_IThe adult orangutan]_I [_Iconstructs a nest every evening out
of leaves and twigs]_I
c. [_IThe adult orangutan]_I [_Iconstructs a nest every evening]_I
[_Iout of leaves and twigs]_I

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For example, (15) can be read with contrastive stress on the word *collect*, as indicated by the italics in the example, but this does not lead to the division of the single *I* into two shorter ones.

- (15) [_IOur neighbors *collect* antique furniture]_I

If, on the other hand, prominence is attributed to a phrase that normally cannot have it, a new intonation contour is created. For example, while pronouns are usually unstressed, under certain circumstances they become semantically prominent and must receive stress. In such cases, the division of *I* constituents must be modified (see Bing, 1979). It should be noted that we are not referring here to contrastive stress, which may appear on pronouns, as on any other type of word, as in *Tell her to wait*. This is the same type of situation as was seen in (15), where contrastive stress does not require a modification of *I* structure, but instead only determines the position of the strong element within *I*. Contrastive prominence refers to a different type of phenomenon. Consider, for example, the sentence in (16a), which is assigned one *I* since it is dominated by a single root sentence. The sentence in (16b) has the same syntactic structure, and thus is also assigned one *I* by the basic *I* formation rule. The presence of the pronouns, however, requires the listener to interpret this sentence in a specific way, which cannot be the same as the interpretation of (16a). The interpretation of (16b) must be expressed phonologically by placing prominence on the pronouns and this, in turn, causes the single *I* to be restructured as three smaller *I*s, as in (16c).

- (16) a. [_IPaul called Paula before Carla called Carl]_I
 b. [_IPaul called Paula before she called him]_I
 c. [_IPaul called Paula]_I [_Ibefore *she*]_I [_Icalled *him*]_I

The different possibilities just seen for restructuring intonational phrases all contribute to the flexibility that has been frequently observed in relation to the domains of intonation contours (see, among others, Selkirk, 1978b, 1984b; Bing, 1979). It is clear that as soon as this type of flexibility is required in delimiting a constituent, the constituent cannot be isomorphic with any syntactic constituent since syntax does not allow for the same type of regrouping of the components of a particular string. In this respect, therefore, the phonological hierarchy differs significantly from the syntactic hierarchy.

The flexibility seen in relation to the construction and restructuring of *I* is not, however, completely free. Since the Strict Layer Hypothesis requires that *I* exhaustively dominate one or more ϕ s, it follows that whenever *I* restructuring occurs, it must occur at the juncture between two

ϕ s. There are, furthermore, certain syntactic constraints on where an *I* may be broken in order to form smaller *Is*. The most important factor seems to be a general tendency to avoid restructuring an *I* in any position other than at the end of a noun phrase. Thus, while the sentence in (17) contains five ϕ s, there are only three likely ways of restructuring the original single *I* formed by the basic *I* construction rule, as seen in (18a-c).

- (17) [_I[The giant panda] _{ϕ} [eats] _{ϕ} [only one type] _{ϕ} [of bamboo] _{ϕ} [in its natural habitat] _{ϕ}]_I
- (18) a. [_I[The giant panda] _{ϕ}]_I [_I[eats] _{ϕ} [only one type] _{ϕ} [of bamboo] _{ϕ}]_I [_I[in its natural habitat] _{ϕ}]_I
 b. [_I[The giant panda] _{ϕ} [eats] _{ϕ} [only one type of bamboo] _{ϕ}]_I [_I[in its natural habitat] _{ϕ}]_I
 c. [_I[The giant panda] _{ϕ}]_I [_I[eats] _{ϕ} [only one type] _{ϕ} [of bamboo] _{ϕ} [in its natural habitat] _{ϕ}]_I

I restructuring is not likely to occur after the verb *eat*, or after the noun *type*, since it is not the end of a noun phrase (i.e. [*only one type of bamboo*]_{NP}).

Further support for the proposal that *I* restructuring occurs at the end of an NP, but not after a noun within an NP, comes from embedded possessive constructions such as that in (19a), and its Italian equivalent in (19b).⁷

- (19) a. [_I[My friend's] _{ϕ} [neighbor's] _{ϕ} [aunt's] _{ϕ} [mother] _{ϕ} [knows] _{ϕ} [a famous writer] _{ϕ}]_I
 b. [_I[La madre] _{ϕ} [della zia] _{ϕ} [del vicino] _{ϕ} [della mia amica] _{ϕ} [conosce] _{ϕ} [una famosa scrittrice] _{ϕ}]_I

The only place in which the long *Is* in these two sentences may be divided is after *mother* in (19a) and after *amica* in (19b). Restructuring after any of the other nouns would interrupt the noun phrase which is the subject of the sentence, and is therefore avoided.

It should be noted that the principle that restructures *Is* after an NP is another source of nonisomorphism between phonological and syntactic constituents in certain cases. That is, restructuring in accordance with the NP restriction may create prosodic structures that are not isomorphic to syntactic structures. Consider the sentence in (20a) and its restructuring in (20b).

- (20) a. [_II would never have believed the children of John and Mary to be able to become so ill-mannered]_I

- b. [_II would never have believed the children of John and Mary]_I
 [_Ito be able to become so ill-mannered]_I

Since the first *I* in (20b) does not represent a syntactic constituent, it is clear that in this case, and in cases like it, the prosodic structure must be different from the syntactic structure.

It should be noted, furthermore, that restructuring is not equally likely to apply after all NPs. Rather, there is a further restriction on dividing an *I* into smaller *I*s at the end of an NP that reflects the argument structure of the predicate in question. That is, there seems to be a general tendency to avoid separating an obligatory argument from its verb even if such a division would respect the NP constraint. Optional arguments, on the other hand, seem not to be affected by the same restriction. Thus, the division of the *I* before the optional argument of the verb *buys* in (21b) is more acceptable than the division before the obligatory argument of *gives* in (22b), despite the fact that the syntactic structure of the two sentences is identical.

- (21) a. [_IThat kind old lady always buys fresh meat for the stray cats
 that live in the park]_I
 b. [_IThat kind old lady always buys fresh meat]_I [_Ifor the stray
 cats that live in the park]_I
- (22) a. [_IThat kind old lady always gives fresh meat to the stray cats
 that live in the park]_I
 b. ?*[_IThat kind old lady always gives fresh meat]_I [_Ito the stray
 cats that live in the park]_I

Thus, it can be concluded that while there is a general preference for breaking an *I* after an NP, this preference may be restricted by the argument structure involved.

In addition to the context following an NP, restructuring of *I* may take place in the context of a new $\bar{S}$. That is, a long *I* may be divided just before the beginning of an $\bar{S}$, but only in cases in which this division does not interrupt an NP. Thus, (23a) may be restructured as (23b), but (24a) may not be restructured as (24b) since doing so would interrupt an NP.

- (23) a. [_II thought you already knew that Gertrude was moving to
 southern Italy]_I
 b. [_II thought you already knew]_I [_Ithat Gertrude was moving to
 southern Italy]_I

- (24) a. [_II thought you knew the family that was moving to southern Italy]_I
 b. *[_II thought you knew the family]_I [_Ithat was moving to southern Italy]_I

These examples show that when the two syntactic factors that affect *I* restructuring – the end of NP and the beginning of $\bar{S}$ – are in conflict, it is the NP that ‘wins’.

There are also cases in which the beginning of $\bar{S}$ may be in conflict with the restriction against separating an obligatory argument from its verb. As the examples in (25) and (26) show, in such cases it is the beginning of $\bar{S}$ that ‘wins’. That is, if necessary, it is possible to override the constraint on restructuring an obligatory argument of a verb, but only if the argument is a sentence. Thus, the division in (25b) is more acceptable than that in (26b).

- (25) a. [_IOur next door neighbor truly believes that black cats bring bad luck]_I
 b. [_IOur next door neighbor truly believes]_I [_Ithat black cats bring bad luck]_I
- (26) a. [_IOur next door neighbor truly believes the myth about black cats and bad luck]_I
 b. ?*[_IOur next door neighbor truly believes]_I [_Ithe myth about black cats and bad luck]_I

A comparison of the three factors that constrain *I* restructuring reveals a type of hierarchy among these factors. The strongest factor is the NP, which may not be broken by restructuring. The second factor in terms of strength seems to be $\bar{S}$, which allows the creation of a new *I*, except in cases in which this would break an NP. The formation of a new *I* at the beginning of $\bar{S}$ may, however, counteract the effect of the third factor, the restriction against separating an internal argument from its verb.

The fact that *I* restructuring tends to take place only in relation to two syntactic contexts shows that in prosodic phonology the constituents NP and $\bar{S}$ play particularly strong roles in determining the final division of a string into *I*s. That NP and $\bar{S}$ behave in a somewhat special way in relation to prosody is not surprising, however, since these are precisely the two cyclic nodes, at least in languages such as English. Assuming that cyclicity is, in fact, a relevant factor in determining the occurrence of *I* restructuring in languages in which other categories are cyclic, the prediction can be made that it will be this set of constituents that is relevant in constraining *I* restructuring (see Nespor and Vogel, 1982). There are, moreover, addition-

al independent reasons in both syntax and semantics for considering NP and $\bar{S}$ somewhat special domains (see, among others, Akmajian, 1975; Bach, 1977; Bing, 1979). The fact that the noun phrase seems to have the stronger effect on *I* restructuring is, furthermore, consistent with observations made by Bing (1979), Gunter (1974), and Ladd (1980), among others, to the effect that nouns are in some sense the most prominent or 'accentable' part of speech.⁸ As far as the role of obligatory arguments in *I* restructuring is concerned, the fact that they tend to inhibit division of a string is a direct reflection of the interaction between argument structure and phonology at the *I* level of the prosodic hierarchy. The fact that the constraint against separating an obligatory argument from its verb is the weakest of the three factors examined here seems to indicate, however, that the effect of syntax on phonology is stronger than the effect of argument structure.

While the vast majority of the cases of *I* restructuring seem to follow the patterns and constraints discussed above, there are some special cases that cannot be accounted for in the same way. It has often been observed that lists and complexly embedded constructions have particular intonation patterns. In fact, the *I* divisions in these cases often violate the general principles proposed above. Consider first the sentences containing lists in (27a-31a) and their possible restructurings in (27b-31b).

- (27) a. [_IThe big fat ugly beast scared away the children]_I
 b. [_IThe big]_I [_Ifat]_I [_Iugly]_I [_Inasty beast]_I
 [scared away the children]_I
- (28) a. [_IThat mountain road is long narrow windy and bumpy]_I
 b. [_IThat mountain road is long]_I [_Inarrow]_I [_Iwindy]_I
 [and bumpy]_I
- (29) a. [_IEveryone at the party ate talked sang and danced]_I
 b. [_IEveryone at the party ate]_I [_Italked]_I [_Isang]_I
 [and danced]_I
- (30) a. [_IDucks geese swans and coots inhabit this lake]_I
 b. [_IDucks]_I [_Igeese]_I [_Iswans]_I [_Iand coots]_I
 [inhabit this lake]_I
- (31) a. [_IThey own two cats three dogs four parakeets and a turtle]_I
 b. [_IThey own two cats]_I [_Ithree dogs]_I [_Ifour parakeets]_I
 [and a turtle]_I

The optional restructuring shown in (27b) violates the NP constraint since it divides the original *I* following APs and not NPs, and in such a way that an NP is broken apart.⁹ The restructurings in (28b) and (29b) also violate the NP constraint since they create new *I*s following APs and VPs, respectively. While the restructurings of the lists in (30b) and (31b) do not violate the NP constraint in the sense that they place the *I* divisions after NPs, they do represent violations of the constraint in the sense that they cause a single (complex) NP to be broken down. Since all of the types of lists follow the same prosodic pattern, it is clear that what is relevant is the fact that they are lists, regardless of the type of constituents involved. A special rule for optional restructuring of lists into separate *I*s seems to be needed in addition to the general principles proposed above. Such a rule can be formulated as follows:

(32) *List Restructuring* (optional)

In a sequence of more than two constituents of the same type, i.e. $x_1, x_2 \dots x_n$, an intonation break may be inserted before each repetition of the node X, i.e. before $x_2, x_3 \dots x_n$.

It should be noted that the word ‘repetition’ in this rule does not refer to the first X in the sequence but to those subsequent nodes that are the same as X.

A less common type of list pattern arises when the list is separated to a greater extent from the matrix sentence, as typically indicated by a colon in orthography. This pattern is found in sentences like those exemplified in (33) and (34).

(33) a. Let’s invite: Arnold, Arthur, Archibald, and Zachary.

b. [_ILet’s invite]_I [_IArnold]_I [_IArthur]_I [_IArchibald]_I
[_Iand Zachary]_I

(34) a. We were told to buy the following: milk, eggs, bread, and cheese

b. [_IWe were told to buy the following]_I [_Imilk]_I [_Ieggs]_I
[_Ibread]_I [_Iand cheese]_I

To handle these cases, it is only necessary to extend the notion of repetition to cover the first X as well as subsequent Xs. In this way, a new *I* is also formed at the beginning of the first instance of X.

Let us now consider embedded structures, the other area in which *I* restructuring may apply in ways that violate the general constraints. Each of the possible restructurings in (35b-37b) of the respective sentences in (35a-37a) violates the general NP constraint on restructuring.

- (35) a. [_IThe book in the bag in the box on the table in the study
belongs to Albert]_I
b. [_IThe book in the bag]_I [_Iin the box]_I [_Ion the table]_I
[_Iin the study]_I [_Ibelongs to Albert]_I
- (36) a. [_IThis is the cat that ate the rat that ate the cheese]_I
b. [_IThis is the cat]_I [_Ithat ate the rat]_I [_Ithat ate the cheese]_I
- (37) a. [_IThe woman that represents the company that owns the stores
that sell the machines that brew coffee automatically is a friend
of mine]_I
b. [_IThe woman]_I [_Ithat represents the company]_I [_Ithat owns the
stores]_I [_Ithat sell the machines]_I [_Ithat brew coffee automati-
cally]_I [_Iis a friend of mine]_I

In each of the restructurings in (35b-37b), an NP is broken into more than one *I*, contrary to the claim made above that *I* restructuring tends to avoid interrupting an NP. It is clear, however, that in embedded structures such as those just given, the only way to divide the original long *I* into shorter *I*s requires that an NP be broken into shorter strings.

Although the embedded structures are syntactically quite different from the lists in (27)-(31), there is also a way in which the two types of constructions are related to each other. Just as the lists contain sequences of the same constituent X, the embedded structures can also be said to contain sequences of a given node X. In (35), the repeated node is PP and in (36) and (37), the repeated node is $\bar{S}$, although these cases differ from the lists in that the repetitions of X begin here before the preceding X ends. It should be noted that in each of the cases in question there is not only a particular node label that recurs several times in the string, but there is also a periodic repetition of the same elements within X.

In some embedded structures, more than one type of node appears in a repeating pattern, and in such cases it is necessary to have a way to decide which of the nodes is to be considered the repeating X for the purposes of *I* restructuring. In the case of (36), in addition to saying that the repeating node is $\bar{S}$, as in (38), it would also be possible to say that the repeating node is NP, as in (39).

(38) This is the cat [$\bar{S}$ that ate the rat [$\bar{S}$ that ate the cheese...

(39) This is [_{NP}the cat that ate [_{NP}the rat that ate [_{NP}the cheese...

As far as the prosodic structure of such sentences is concerned, however, there is not a choice. The division into *I*s can only be that seen in (36),

the one that corresponds to the repetition pattern in (38). Once again, the noun seems to play a fundamental role in prosody, since in the cases in which more than one type of X may be taken as the repeating node, the division into shorter *Is* is always such that the *Is* end in a noun.

Before concluding our examination of restructuring of embedded constructions, it must be pointed out that not all cases can be handled as repetitions. For example, the sentence in (40) has a complex embedded structure, although the pattern of repeating nodes is not as regular as it is in the embedded structures seen above.

- (40) [_I I know the artist that painted the picture of the woman that wrote the book that won the acclaim of many]_I

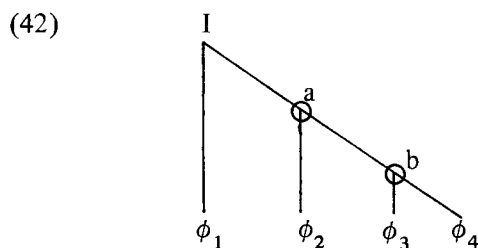
While there are three instances of $\bar{S}$ in this sentence, we cannot count them as a list, as was done in the other embedded structures, since in (40) there is not always a periodic repetition of the same elements in each case. In fact, in such cases, it is much more difficult to decide where the single long *I* corresponding to the root sentence may be broken down into shorter *Is*. The results of a division on the basis of the three $\bar{S}$ s is quite unacceptable, as can be seen in (41).

- (41) * [_I I know the artist]_I [_I that painted the picture of the woman]_I
 [_I that wrote the book]_I [_I that won the acclaim of many]_I

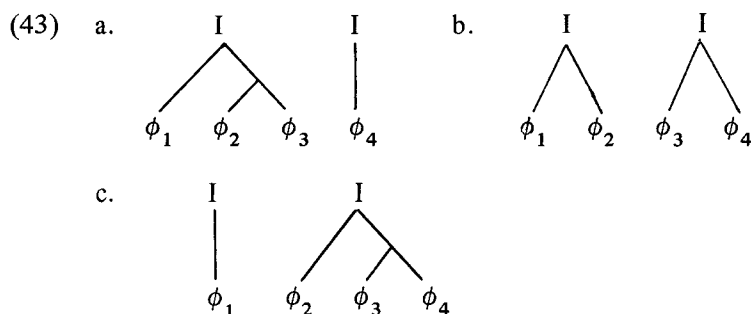
What lists and embedded structures show is that besides the basic principles that govern *I* restructuring there are a few specific cases in which additional principles are needed. These additional principles are not, however, totally unrelated to those that account for the rest of the language. That is, in the cases of lists and embedded structures that behave like lists, we have seen that the notion of repetition is crucial. The prosodic effect of this repetition is a series of *Is* that are of approximately the same length. A similar principle was seen earlier in relation to the division of more typical long *Is* into several shorter ones, where it was observed that there is a general tendency to restructure *Is* such that the result is a series of *Is* of more or less the same length. These principles, along with the interacting factor of speech rate, seem to reflect a general abstract rhythmic organization of speech into chunks of a more or less fixed temporal length. In addition, in the case of embedded constructions, in regular repetitions as well as others, it was observed that the noun plays a central role in determining where intonation breaks may occur. While in embedded constructions we cannot follow the principle used in other cases of restructuring – that is, the introduction of an intonation break at the end of an NP – the category N still seems to be relevant in restruc-

turing, additional evidence of the special role the noun plays in prosody.

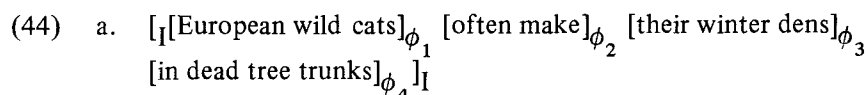
Finally, it should be noted that *I* restructuring provides additional motivation for the n-ary branching trees proposed here, as opposed to the binary branching trees proposed in earlier work on prosodic phonology (see also section 1.2.1 above). In a binary branching structure, intermediate nodes are often formed between a given node of level X^n , in the case *I*, and the nodes of level X^{n-1} , in this case ϕ , as seen in (42).



The additional nodes, indicated by the circles in (42), represent additional constituent structure as well. That is, node *a* dominates a constituent consisting of $\phi_2 - \phi_4$, and node *b* dominates a constituent consisting of $\phi_3 - \phi_4$. The presence of such constituents internal to *I* would seem to indicate, further, that this structure should be respected during *I* restructuring. This was in fact proposed in Nespor and Vogel (1982), where it was suggested that *I* restructuring was possible only when the new *I*s that split off from the larger one represented a constituent in the original tree. Only in special cases of emphasis was some other division allowed. The predictions made in relation to (42) by such a claim are that the *I* consisting of $\phi_1 - \phi_4$ could be broken down only as shown in (43).



A new *I* could not be created from ϕ_2 and ϕ_3 . There are, however, cases in which precisely this combination represents the correct restructuring, as seen in (44).



- b. $[I[\text{European wild cats}]_{\phi_1}]_I [I[\text{often make}]_{\phi_2}]_I$
 $[\text{their winter dens}]_{\phi_3}]_I [I[\text{in dead tree trunks}]_{\phi_4}]_I$

A similar problem also arises with (18a) above, where restructuring of five ϕ s creates a new I from $\phi_2 - \phi_4$, a string which would not form a constituent in a binary branching tree. These examples, and many others like them, cannot be accounted for on the grounds that they represent cases of special emphasis, as was originally proposed, since in fact they represent quite neutral pronunciations. It would, of course, be possible to ignore the internal constituent structure in I restructuring, but doing so would destroy one of the functions of precisely this structure.

The use of n-ary branching trees avoids the problems just seen in relation to binary branching trees. That is, since there is no additional constituent structure between I and ϕ , restructuring may group any sequence of ϕ s into smaller I s, as long as the division respects the syntactic and argument structure conditions as well as the general timing conditions discussed above. The proposed restructurings in (44) and (18a) respect these conditions.

Thus far, we have only examined the intonational phrase from the point of view of intonation contours, potential pause positions, and relative prominence patterns. While these phenomena would be adequate in themselves as motivation for the I constituent in phonology, there are still other phenomena that operate within I . Specifically, in the following sections, it will be demonstrated that there are also segmental phonological rules that have I as their domain of application.

7.3. Segmental rules in the intonational phrase

For every prosodic constituent we have examined thus far, we have found what we are calling prosodic rules in a variety of languages. The constituent I is no exception, as will be demonstrated below on the basis of several rules in Italian, Spanish, and Greek. Since I is a fairly flexible entity, it follows that the domains of application of I -level rules must also be flexible. That this is in fact the case will also be shown in relation to each of the rules discussed below.

7.3.1. Italian

The first sandhi rule we will examine here is Gorgia Toscana (GT), a rule that is found in the Tuscan variety of Italian. As was mentioned earlier (section 2.3.1), this rule essentially changes the voiceless stops /p, t, k/ into the corresponding fricatives [ϕ , θ , h] between two [–consonantal] segments within and across words.¹⁰ On the basis of our recordings of over 900 sentences read by five speakers of Tuscan Italian from Florence